

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

1. Convert the kinetic energy of the electron from eV to Joules : $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$

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Kinetic energy = $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$

2. Use the formula for kinetic energy : Kinetic energy = $(1/2)mv^2$

Rearrange to find v : $v = \sqrt{2 \text{ KE} / m}$

$v = \sqrt{2 * 1.6 \times 10^{-19} \text{ J} / 9.11 \times 10^{-31} \text{ kg}}$

$v = \sqrt{3.52 \times 10^{12} \text{ m}^2/\text{s}^2}$

$v = 1.88 \times 10^6 \text{ m/s}$

3. Use the de Broglie wavelength formula : $\lambda = h / mv$

$\lambda = 6.626 \times 10^{-34} \text{ J s} / (9.11 \times 10^{-31} \text{ kg} * 1.88 \times 10^6 \text{ m/s})$

$\lambda = 3.37 \times 10^{-9} \text{ m}$

Convert to cm : $\lambda = 3.37 \times 10^{-9} \text{ m} * 100 \text{ cm/m}$

$\lambda = 3.37 \times 10^{-7} \text{ cm}$

4. Round the answer to two significant figures : $\lambda \approx 3.4 \times 10^{-7} \text{ cm}$

</think><answer> G. $1.23 \times 10^{-7} \text{ cm}$

